

ASSIGNMENT

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?
 - Native mobile application development creates apps specifically for Android or iOS with the best performance and full access to device features, hybrid development uses web technologies to build apps that run on multiple platforms but may have lower performance, while cross-platform development uses frameworks like Flutter to build apps for both Android and iOS from a single codebase with near-native performance, making each approach suitable depending on the project's requirements, budget, and target users.
2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?
 - The Flutter Ecosystem is a complete set of tools, libraries, widgets, plugins, and the Dart programming language that enables developers to create modern, high-performance mobile applications for Android and iOS from a single codebase, helping reduce development time while maintaining a consistent user experience.
3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?
 - The Software Development Life Cycle (SDLC) is a structured process that includes planning, designing, developing, testing, deploying, and maintaining software, and it is important because it helps developers produce organized, reliable, and high-quality applications that meet user needs while reducing errors and project risks.
4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?
 - Flutter development practices and the SDLC work together by providing a structured workflow for planning, designing, building, testing, deploying, and maintaining applications, while Flutter's features such as a single codebase, reusable widgets, and Hot Reload make each stage of development faster and more efficient.
5. Why should developers understand these concepts before starting an application project?
 - Developers should understand these concepts before starting an application project because they help in selecting the most appropriate development approach, planning the project effectively, avoiding common mistakes, improving software quality, reducing development time and costs, and delivering applications that meet user and business requirements.